Proofs

1 Proof of slack concentration

Proposition 1.1 (slack concentration). *For a fixed first-level subgraph $G^{(s)}$ and a fixed demand subset $B' \subseteq B^{(s)}$, let $N(B') = \left\{a \in A^{(s)} : \exists\, b \in B',\ M_{ab} = 1\right\}$ denote the supplies eligible for at least one demand in B'. Define the slack of B' in $G^{(s)}$ as*

$$\operatorname{slack}^{(s)}(B') = \sum_{a \in N(B')} \operatorname{capacity}(a) - \sum_{b \in B'} \operatorname{demand}(b).$$

For each $i \in \{1, \dots, m\}$, the corresponding local slack in $G^{(s,i)}$ is

$$\operatorname{slack}^{(s,i)}(B') = \sum_{a \in N(B') \cap A^{(s,i)}} \operatorname{capacity}(a) - \frac{1}{m} \sum_{b \in B'} \operatorname{demand}(b).$$

Under the supply partition described above, for every $i \in \{1, \dots, m\}$,

$$\mathbb{E}\left[\operatorname{slack}^{(s,i)}(B')\right] = \frac{\operatorname{slack}^{(s)}(B')}{m}.$$

Moreover, if $\operatorname{slack}^{(s)}(B') > 0$, then

$$\Pr\left[\operatorname{slack}^{(s,i)}(B') < 0\right] \le \exp\left(-\frac{2\operatorname{slack}^{(s)}(B')^2}{m^2 \sum_{a \in N(B')} \operatorname{capacity}(a)^2}\right).$$

Proof. Fix $i \in \{1, \dots, m\}$. For each $a \in N(B')$, let

$$Z_a = \begin{cases} 1, & a \in A^{(s,i)}, \\ 0, & a \notin A^{(s,i)}. \end{cases}$$

Since $A^{(s,i)}$ is sampled uniformly without replacement, $\mathbb{E}[Z_a] = 1/m$. Therefore,

$$\begin{aligned} \mathbb{E}\left[\operatorname{slack}^{(s,i)}(B')\right] &= \sum_{a \in N(B')} \operatorname{capacity}(a)\mathbb{E}[Z_a] - \frac{1}{m} \sum_{b \in B'} \operatorname{demand}(b) \\ &= \frac{\operatorname{slack}^{(s)}(B')}{m}. \end{aligned}$$

The global and local slack definitions further give

$$\operatorname{slack}^{(s,i)}(B') - \frac{\operatorname{slack}^{(s)}(B')}{m} = \sum_{a \in N(B')} \operatorname{capacity}(a)\left(Z_a - \frac{1}{m}\right).$$

The variables $\{Z_a\}_{a \in N(B')}$ are negatively associated, and each summand on the right-hand side has zero expectation and a range of length $\operatorname{capacity}(a)$. Hence, if $\operatorname{slack}^{(s)}(B') > 0$, Hoeffding's inequality gives

$$\begin{aligned} &\Pr\left[\operatorname{slack}^{(s,i)}(B') < 0\right] \\ &= \Pr\left[\begin{array}{c} \displaystyle\sum_{a \in N(B')} \operatorname{capacity}(a)\left(Z_a - \frac{1}{m}\right) \\ < -\dfrac{\operatorname{slack}^{(s)}(B')}{m} \end{array}\right] \\ &\le \exp\left(-\frac{2\operatorname{slack}^{(s)}(B')^2}{m^2 \sum_{a \in N(B')} \operatorname{capacity}(a)^2}\right). \end{aligned}$$

□

2 Proof of Expected relative underdelivery

Proposition 2.1 (Expected relative underdelivery). *Suppose that $G^{(s)}$ has an allocation $x = (x_{ab})_{(a,b) \in E^{(s)}}$ that exactly satisfies every demand. For this allocation, define $L \ge 1$ such that $1/L$ is the largest fraction of any demand contributed by a single supply:*

$$\frac{1}{L} = \max_{a \in A^{(s)},\, b \in B^{(s)}} \frac{M_{ab} x_{ab}}{\operatorname{demand}(b)}.$$

Under the randomized partition of the supply set $A^{(s)}$, define the underdelivery of demand b in subproblem $G^{(s,i)}$ as

$$u_b^{(s,i)} = \left[\frac{\text{demand}(b)}{m} - \sum_{a \in A^{(s,i)}} M_{ab} x_{ab} \right]_+ .$$

Then the expected relative underdelivery after partitioning satisfies

$$\frac{\mathbb{E}\left[\sum_{i=1}^{m} \sum_{b \in B^{(s)}} u_b^{(s,i)}\right]}{\sum_{b \in B^{(s)}} \text{demand}(b)} \le \frac{1}{2}\sqrt{\frac{m-1}{L}}.$$

Proof. Fix $b \in B^{(s)}$ and $i \in \{1, \dots, m\}$. Since every supply is assigned to $A^{(s,i)}$ with probability $1/m$ and x exactly satisfies demand b,

$$\begin{aligned} \mathbb{E}\left[\sum_{a \in A^{(s,i)}} M_{ab} x_{ab} \right] &= \frac{1}{m} \sum_{a \in A^{(s)}} M_{ab} x_{ab} \\ &= \frac{\text{demand}(b)}{m}. \end{aligned}$$

Because $A^{(s,i)}$ is a uniformly random subset of fixed size, the inclusion events of different supplies have nonpositive covariances. Therefore,

$$\text{Var}\left[\sum_{a \in A^{(s,i)}} M_{ab} x_{ab} \right] \le \frac{m-1}{m^2} \sum_{a \in A^{(s)}} (M_{ab} x_{ab})^2 .$$

Thus, the expected allocation to b in each subproblem equals its assigned demand. Its expected excess above $\text{demand}(b)/m$ therefore equals its expected underdelivery below this value. Since their sum is the absolute deviation, the definition of $u_b^{(s,i)}$ gives

$$\begin{aligned} \mathbb{E}\left[u_b^{(s,i)}\right] &= \frac{1}{2}\mathbb{E}\left[\left| \sum_{a \in A^{(s,i)}} M_{ab} x_{ab} - \frac{\text{demand}(b)}{m} \right| \right] \\ &\le \frac{1}{2}\sqrt{\text{Var}\left[\sum_{a \in A^{(s,i)}} M_{ab} x_{ab} \right]} \\ &\le \frac{\sqrt{m-1}}{2m} \sqrt{\sum_{a \in A^{(s)}} (M_{ab} x_{ab})^2}, \end{aligned}$$

where the first inequality follows from the Cauchy–Schwarz inequality. By the definition of L, each supply contributes at most $\text{demand}(b)/L$ to demand b. Hence,

$$\begin{aligned} \sum_{a \in A^{(s)}} (M_{ab} x_{ab})^2 &\le \frac{\text{demand}(b)}{L} \sum_{a \in A^{(s)}} M_{ab} x_{ab} \\ &= \frac{\text{demand}(b)^2}{L}. \end{aligned}$$

Combining the preceding bounds gives

$$\mathbb{E}\left[u_b^{(s,i)}\right] \le \frac{\text{demand}(b)}{2m}\sqrt{\frac{m-1}{L}}.$$

Summing over all $i \in \{1, \dots, m\}$ and $b \in B^{(s)}$ yields

$$\mathbb{E}\left[\sum_{i=1}^{m} \sum_{b \in B^{(s)}} u_b^{(s,i)} \right] \le \frac{1}{2}\sqrt{\frac{m-1}{L}} \sum_{b \in B^{(s)}} \text{demand}(b).$$

Dividing both sides by the total demand proves the result. □

3 Proof of Demand Adjustment Phase

Proposition 3.1 (approximation guarantee for demand adjustment). *Let Δ denote the total deficit reduction achieved by the demand adjustment phase. Let $\Delta^\star$ denote the maximum total deficit reduction achievable from the same initial allocation by applying admissible reallocation chains. Then*

$$\Delta \ge \frac{1}{3}\Delta^\star.$$

Proof. For $a \in A$ and $b \in B$, let r_a^0 and x_{ab}^0 denote the values of r_a and x_{ab} at the beginning of the demand adjustment phase, respectively, and let r_a and x_{ab} denote their values at the end of the phase. For $b \in D^{\text{unsat}}$, let q_b^0 and q_b denote the deficit of b at the beginning and end of the phase, respectively. Let $\mathcal{P}$ be the candidate set of all reallocation chains admissible. For each $P \in \mathcal{P}$, write

$$P = (a_0^P, b_0^P, a_1^P, b_1^P).$$

Suppose that the demand adjustment phase successively applies $P_1, \ldots, P_T \in \mathcal{P}$. Its total reallocation amount is

$$\Delta = \sum_{t=1}^{T} \delta(P_t).$$

For each $P \in \mathcal{P}$, let $\Delta_P \geq 0$ denote the amount reallocated along P. The maximum feasible reallocation amount $\Delta^\star$ is defined by

$$\Delta^\star = \max_{\{\Delta_P \geq 0\}_{P \in \mathcal{P}}} \sum_{P \in \mathcal{P}} \Delta_P,$$

subject to

$$\sum_{\substack{P \in \mathcal{P} \\ a_0^P = a}} \Delta_P \leq r_a^0, \qquad \forall a \in S^{\text{rem}},$$

$$\sum_{\substack{P \in \mathcal{P} \\ (a_1^P, b_0^P) = (a,b)}} \Delta_P \leq x_{ab}^0, \qquad \forall a \in S^{\text{adj}},\ b \in D^{\text{sat}},$$

$$\sum_{\substack{P \in \mathcal{P} \\ b_1^P = b}} \Delta_P \leq q_b^0, \qquad \forall b \in D^{\text{unsat}}.$$

Let $\{\Delta_P^\star\}_{P \in \mathcal{P}}$ be an optimal solution and define

$$\mathcal{P}^\star = \left\{ P \in \mathcal{P} \;\middle|\; \Delta_P^\star > 0 \right\}.$$

Thus, $\mathcal{P}^\star$ is the set of chains used by the optimal reallocation. Moreover,

$$\Delta^\star = \sum_{P \in \mathcal{P}^\star} \Delta_P^\star.$$

At the end of the demand adjustment phase, no admissible chain remains. Therefore, every $P \in \mathcal{P}^\star$ satisfies

$$r_{a_0^P} = 0 \quad \text{or} \quad x_{a_1^P b_0^P} = 0 \quad \text{or} \quad q_{b_1^P} = 0.$$

To obtain a disjoint partition of $\mathcal{P}^\star$, define

$$\mathcal{P}_r = \left\{ P \in \mathcal{P}^\star \;\middle|\; r_{a_0^P} = 0 \right\},$$

$$\mathcal{P}_x = \left\{ P \in \mathcal{P}^\star \;\middle|\; r_{a_0^P} > 0,\ x_{a_1^P b_0^P} = 0 \right\},$$

and

$$\mathcal{P}_q = \left\{ P \in \mathcal{P}^\star \;\middle|\; r_{a_0^P} > 0,\ x_{a_1^P b_0^P} > 0,\ q_{b_1^P} = 0 \right\}.$$

The termination condition ensures that these sets are pairwise disjoint and satisfy

$$\mathcal{P}^\star = \mathcal{P}_r \cup \mathcal{P}_x \cup \mathcal{P}_q.$$

Consequently,

$$\Delta^\star = \sum_{P \in \mathcal{P}_r} \Delta_P^\star + \sum_{P \in \mathcal{P}_x} \Delta_P^\star + \sum_{P \in \mathcal{P}_q} \Delta_P^\star.$$

It follows from the definition of $\Delta^\star$ that

$$\sum_{P \in \mathcal{P}_r} \Delta_P^\star \leq \sum_{\substack{a \in S^{\text{rem}} \\ r_a = 0}} r_a^0,$$

$$\sum_{P \in \mathcal{P}_x} \Delta_P^\star \leq \sum_{\substack{a \in S^{\text{adj}},\ b \in D^{\text{sat}} \\ x_{ab} = 0}} x_{ab}^0,$$

and

$$\sum_{P \in \mathcal{P}_q} \Delta_P^\star \leq \sum_{\substack{b \in D^{\text{unsat}} \\ q_b = 0}} q_b^0.$$

Each reallocation of amount $\delta(P_t)$ decreases one residual capacity in S^{rem}, one allocation from S^{adj} to D^{sat}, and one deficit in D^{unsat} by exactly $\delta(P_t)$. Summing over all applied chains gives

$$\sum_{a \in S^{\text{rem}}} (r_a^0 - r_a) = \Delta,$$
$$\sum_{a \in S^{\text{adj}},\, b \in D^{\text{sat}}} (x_{ab}^0 - x_{ab}) = \Delta,$$
$$\sum_{b \in D^{\text{unsat}}} (q_b^0 - q_b) = \Delta.$$

Restricting these sums to quantities exhausted at the end of the phase yields

$$\sum_{\substack{a \in S^{\text{rem}} \\ r_a = 0}} r_a^0 \le \Delta,$$

$$\sum_{\substack{a \in S^{\text{adj}},\, b \in D^{\text{sat}} \\ x_{ab} = 0}} x_{ab}^0 \le \Delta,$$

and

$$\sum_{\substack{b \in D^{\text{unsat}} \\ q_b = 0}} q_b^0 \le \Delta.$$

Therefore, each of the three terms in the decomposition of $\Delta^\star$ is at most Δ, and hence

$$\begin{aligned}\Delta^\star &= \sum_{P \in \mathcal{P}_r} \Delta_P^\star + \sum_{P \in \mathcal{P}_x} \Delta_P^\star + \sum_{P \in \mathcal{P}_q} \Delta_P^\star \\ &\le 3\Delta.\end{aligned}$$

Equivalently,

$$\Delta \ge \frac{1}{3}\Delta^\star.$$

□

4 Proof of heat replacement Phase

Proposition 4.1 (local optimality of heat replacement). *At the end of the heat replacement phase, for every heat replacement P and every $0 < \eta \le \delta(P)$, reallocating η units along P cannot improve the offline objective.*

Proof. At termination, consider any heat replacement $P = (a_0, b, a_1)$. For $0 \le \eta \le \delta(P)$, let $g_P(\eta)$ denote the objective change obtained by reallocating η units along P. Then

$$g_P(\eta) = 2\eta\big(w_{a_1} r_{a_1} - w_{a_0} r_{a_0}\big) + \eta^2\big(w_{a_0} + w_{a_1}\big).$$

Since $w_{a_0} + w_{a_1} > 0$, $g_P(\eta)$ is convex on $[0, \delta(P)]$. Moreover,

$$g_P(0) = 0.$$

Since no admissible replacement remains at termination,

$$g_P(\delta(P)) = \Delta f(P) \le 0.$$

Therefore, for every $0 \le \eta \le \delta(P)$,

$$g_P(\eta) \le \frac{\eta}{\delta(P)} g_P(\delta(P)) \le 0.$$

Thus, no candidate P admits a feasible replacement amount that improves the offline objective after the heat replacement phase. □

5 Proof of Cross-Subproblem Recovery

Proposition 5.1 (total-allocation guarantee after cross-subproblem recovery). *Suppose that the direct allocation step and the subsequent global demand adjustment phase are both run to termination. Let Δ denote the total allocation in the final solution obtained on $G^{(s)}$ after cross-subproblem recovery, and let $\Delta^\star$ denote the maximum total allocation achievable by any feasible allocation on $G^{(s)}$ that contains no overdelivery. Then*

$$\Delta \ge \frac{2}{3}\Delta^\star.$$

Proof. Let

$$x^0 = \{x^0_{ab}\}_{(a,b)\in E^{(s)}} \qquad \text{and} \qquad x = \{x_{ab}\}_{(a,b)\in E^{(s)}}$$

denote the allocations immediately after the direct allocation step in cross-subproblem recovery and at the end of the subsequent global demand adjustment phase, respectively. For each $a \in A^{(s)}$ and $b \in B^{(s)}$, let r^0_a and q^0_b denote the residual capacity and demand deficit associated with x^0, and let r_a and q_b denote their values associated with x.

At the beginning of the global demand adjustment phase, define the following fixed sets:

$$\begin{aligned}
D^{\text{unsat}}_{\text{G}} &= \{b \in B^{(s)} : q^0_b > 0\},\\
D^{\text{sat}}_{\text{G}} &= B^{(s)} \setminus D^{\text{unsat}}_{\text{G}},\\
S^{\text{rem}}_{\text{G}} &= \{a \in A^{(s)} : r^0_a > 0\},\\
S^{\text{adj}}_{\text{G}} &= \left\{a \in A^{(s)} : r^0_a = 0,\ \exists b \in D^{\text{sat}}_{\text{G}} \text{ such that } x^0_{ab} > 0\right\}.
\end{aligned}$$

These sets remain fixed throughout the global demand adjustment phase. During this phase, a reallocation chain

$$P = (a_0, b_0, a_1, b_1) \in S^{\text{rem}}_{\text{G}} \times D^{\text{sat}}_{\text{G}} \times S^{\text{adj}}_{\text{G}} \times D^{\text{unsat}}_{\text{G}}$$

is admissible if

$$(a_0, b_0), (a_1, b_0), (a_1, b_1) \in E^{(s)}$$

and

$$r_{a_0} > 0, \qquad x_{a_1 b_0} > 0, \qquad q_{b_1} > 0.$$

The global demand adjustment phase terminates when no admissible reallocation chain remains. At the termination of the direct allocation step,

$$r^0_a = 0 \quad \text{or} \quad q^0_b = 0, \qquad \forall (a,b) \in E^{(s)}.$$

During global demand adjustment, residual capacities and demand deficits do not increase. Therefore,

$$r^0_a \geq r_a, \qquad \forall a \in A^{(s)},$$

and

$$q^0_b \geq q_b, \qquad \forall b \in B^{(s)}.$$

Moreover, only supplies in $S^{\text{rem}}_{\text{G}}$ can receive additional allocation to demands in $D^{\text{sat}}_{\text{G}}$. Hence,

$$x^0_{ab} \geq x_{ab}, \qquad a \notin S^{\text{rem}}_{\text{G}}, \quad b \in D^{\text{sat}}_{\text{G}}.$$

By the definition of Δ,

$$\Delta = \sum_{a\in A^{(s)}} \sum_{b\in B^{(s)}} M_{ab} x_{ab}.$$

Let

$$x^\star = \{x^\star_{ab}\}_{(a,b)\in E^{(s)}}$$

be a feasible allocation attaining the maximum total allocation on $G^{(s)}$. Then

$$\Delta^\star = \sum_{a\in A^{(s)}} \sum_{b\in B^{(s)}} M_{ab} x^\star_{ab}.$$

By construction, x contains no overdelivery, and $x^\star$ contains no overdelivery by the definition of $\Delta^\star$. We next characterize the additional allocation in $x^\star$ relative to x. The difference between $x^\star$ and x admits the standard alternating decomposition into simple paths that increase the total allocation and components that preserve the total allocation. The latter components can be discarded. Let $\mathcal{C}$ denote the remaining alternating reallocation paths. For each $C \in \mathcal{C}$, write

$$C = (a^C_0, b^C_0, a^C_1, b^C_1, \ldots, a^C_{\ell_C}, b^C_{\ell_C}),$$

where $\ell_C \geq 0$ is the number of allocation edges from which allocation is removed. Such a path satisfies

$$\begin{aligned}
&(a^C_i, b^C_i) \in E^{(s)}, \qquad i = 0, \ldots, \ell_C,\\
&(a^C_i, b^C_{i-1}) \in E^{(s)}, \qquad x_{a^C_i b^C_{i-1}} > 0, \qquad i = 1, \ldots, \ell_C,
\end{aligned}$$

and

$$r_{a^C_0} > 0, \qquad q_{b^C_{\ell_C}} > 0.$$

For each $C \in \mathcal{C}$, let $\Delta_C > 0$ denote the amount associated with C in this decomposition. Then

$$\Delta^\star - \Delta = \sum_{C\in\mathcal{C}} \Delta_C.$$

Moreover, the total amount removed from any allocation edge cannot exceed its current allocation. Therefore,

$$\sum_{\substack{C\in\mathcal{C},\ 1\le i\le \ell_C \\ (a_i^C,b_{i-1}^C)=(a,b)}} \Delta_C \le x_{ab}, \qquad \forall (a,b)\in E^{(s)}.$$

We next show that every path in C removes allocation from at least two edges. Suppose first that $\ell_C = 0$. Then

$$C = (a_0^C, b_0^C),$$

where

$$(a_0^C, b_0^C) \in E^{(s)}, \qquad r_{a_0^C} > 0, \qquad q_{b_0^C} > 0.$$

Since residual capacities and demand deficits do not increase,

$$r^0_{a_0^C} > 0, \qquad q^0_{b_0^C} > 0,$$

contradicting the termination condition of the direct allocation step. Therefore, $\ell_C \neq 0$. Suppose next that $\ell_C = 1$. Write

$$C = (a_0, b_0, a_1, b_1).$$

By the definition of an alternating reallocation path,

$$(a_0, b_0), (a_1, b_0), (a_1, b_1) \in E^{(s)}$$

and

$$r_{a_0} > 0, \qquad x_{a_1 b_0} > 0, \qquad q_{b_1} > 0.$$

Since residual capacities and demand deficits do not increase,

$$r^0_{a_0} > 0, \qquad q^0_{b_1} > 0.$$

Therefore,

$$a_0 \in S_{\mathrm{G}}^{\mathrm{rem}} \qquad \text{and} \qquad b_1 \in D_{\mathrm{G}}^{\mathrm{unsat}}.$$

Since $(a_0, b_0) \in E^{(s)}$ and $r^0_{a_0} > 0$, the termination condition of the direct allocation step gives

$$q^0_{b_0} = 0,$$

and hence

$$b_0 \in D_{\mathrm{G}}^{\mathrm{sat}}.$$

Similarly, since $(a_1, b_1) \in E^{(s)}$ and $q^0_{b_1} > 0$, we have

$$r^0_{a_1} = 0.$$

Thus, $a_1 \notin S_{\mathrm{G}}^{\mathrm{rem}}$. Since only supplies in $S_{\mathrm{G}}^{\mathrm{rem}}$ can receive additional allocation to demands in $D_{\mathrm{G}}^{\mathrm{sat}}$,

$$x^0_{a_1 b_0} \ge x_{a_1 b_0} > 0.$$

Together with $r^0_{a_1} = 0$ and $b_0 \in D_{\mathrm{G}}^{\mathrm{sat}}$, this implies

$$a_1 \in S_{\mathrm{G}}^{\mathrm{adj}}.$$

Consequently,

$$(a_0, b_0, a_1, b_1) \in S_{\mathrm{G}}^{\mathrm{rem}} \times D_{\mathrm{G}}^{\mathrm{sat}} \times S_{\mathrm{G}}^{\mathrm{adj}} \times D_{\mathrm{G}}^{\mathrm{unsat}}.$$

At the end of the global demand adjustment phase, it still satisfies

$$r_{a_0} > 0, \qquad x_{a_1 b_0} > 0, \qquad q_{b_1} > 0.$$

Thus,

$$P = (a_0, b_0, a_1, b_1)$$

remains an admissible reallocation chain, contradicting the termination condition of the global demand adjustment phase. Therefore, $\ell_C \neq 1$. Hence, every path in C satisfies $\ell_C \geq 2$. It follows that

$$\begin{aligned}
2(\Delta^\star - \Delta) &= 2 \sum_{C \in \mathcal{C}} \Delta_C \\
&\leq \sum_{C \in \mathcal{C}} \ell_C \Delta_C \\
&= \sum_{(a,b) \in E^{(s)}} \sum_{\substack{C \in \mathcal{C},\ 1 \leq i \leq \ell_C \\ (a_i^C, b_{i-1}^C) = (a,b)}} \Delta_C \\
&\leq \sum_{(a,b) \in E^{(s)}} x_{ab} \\
&= \sum_{a \in A^{(s)}} \sum_{b \in B^{(s)}} M_{ab} x_{ab} \\
&= \Delta.
\end{aligned}$$

Therefore,

$$2\Delta^\star \leq 3\Delta,$$

or equivalently,

$$\Delta \geq \frac{2}{3}\Delta^\star.$$

□

6 Proof of Incremental Allocation

Proposition 6.1 (incremental allocation guarantee relative to reoptimization). *Let $b_1^{\text{on}}, \ldots, b_t^{\text{on}}$ denote the first t online contracts indexed by their arrival order, and define $B_t = B^{\text{off}} \cup \{b_1^{\text{on}}, \ldots, b_t^{\text{on}}\}$. Let A_t denote the total allocation maintained after processing these online contracts, and let $A_t^\star$ denote the maximum total allocation obtainable by re-solving the entire allocation problem for all contracts in B_t without overdelivering any offline contract. For each $\tau \leq t$, let R_τ denote the maximum amount that can still be allocated to b_τ^{on} through the admissible chains remaining when its processing stops. Then*

$$A_t \geq \frac{2}{3}\left(A_t^\star - \sum_{\tau=1}^{t} R_\tau\right).$$

In particular, if every arrival is processed until no admissible chain remains, then

$$A_t \geq \frac{2}{3}A_t^\star.$$

Proof. For each $b \in B^{\text{off}}$, let u_b^{off} denote its underdelivery in the offline allocation. For each $\tau \leq t$, let V_τ^{on} denote the allocation received by b_τ^{on}. Since the offline allocation contains no overdelivery and online reallocations preserve the total allocation of every existing contract,

$$A_t = \sum_{b \in B^{\text{off}}} \left(\text{demand}(b) - u_b^{\text{off}}\right) + \sum_{\tau=1}^{t} V_\tau^{\text{on}}.$$

$A_t^\star$ is given by

$$\max \sum_{a \in A} \sum_{b \in B_t} M_{ab} x_{ab}$$

subject to

$$\sum_{b \in B_t} M_{ab} x_{ab} \leq \text{capacity}(a), \qquad \forall a \in A,$$

$$\sum_{a \in A} M_{ab} x_{ab} + u_b = \text{demand}(b), \qquad \forall b \in B^{\text{off}},$$

Consider the alternating reallocation paths defined in the proof of Proposition **??**. For each such path C, let ℓ_C denote the number of current allocation edges from which allocation is removed. At the end of the offline stage, no path with $\ell_C \leq 1$ remains by the termination conditions established in Proposition **??**.

When b_τ^{on} arrives, the completed direct allocation step exhausts the residual capacity of every supply adjacent to b_τ^{on}. Therefore, no path with $\ell_C = 0$ can end at b_τ^{on}. A path with $\ell_C = 1$ ending at b_τ^{on} has the form

$$C = (a_0, b_0, a_1, b_\tau^{\text{on}}),$$

where b_0 is an existing contract, $r_{a_0} > 0$, and $x_{a_1 b_0} > 0$. By the definitions of S^{adj} and S^{rem} for the current arrival, such a path is exactly an admissible online reallocation chain. Hence, when the processing of b_τ^{on} stops, the maximum total augmentation obtainable through all remaining length-one paths ending at b_τ^{on} is R_τ.

We next show that subsequent arrivals cannot create or increase the augmentation available through a path with $\ell_C \le 1$ ending at an earlier contract. Since residual capacities never increase, no new path with $\ell_C = 0$ can be created.

First, consider the direct allocation for a later contract b_σ^{on}, where $\sigma > \tau$. Such an allocation increases only an edge $(a, b_\sigma^{\text{on}})$ for which a has positive residual capacity immediately before the update. Suppose that this edge creates or increases a length-one path ending at an earlier contract b. Then a must also be eligible for b. Since residual capacities never increase, a already had positive residual capacity when the processing of b stopped. If b is an online contract, this contradicts the completion of its direct allocation step. If b is an offline contract, this contradicts the absence of a path with $\ell_C = 0$ at the end of the offline stage. Therefore, a subsequent direct allocation cannot create or increase a length-one path ending at an earlier contract.

Next, consider a subsequent online reallocation chain

$$P = (a_0, b_0, a_1, b_\sigma^{\text{on}}).$$

The only allocation edges increased by this chain are (a_0, b_0) and $(a_1, b_\sigma^{\text{on}})$. If (a_0, b_0) creates or increases a length-one path ending at an earlier contract b, then a_0 must be eligible for b. Since $r_{a_0} > 0$ before applying P, a path with $\ell_C = 0$ ending at b would already have existed when b was processed, which is a contradiction. The edge $(a_1, b_\sigma^{\text{on}})$ also cannot create or increase such a path, because this would require a residual supply adjacent to b_σ^{on}, whereas the completed direct allocation step for b_σ^{on} exhausted all such residual capacities. All other allocation edges are unchanged or decreased.

Consequently, every length-one path remaining after the first t arrivals and ending at b_τ^{on} was already available when the processing of b_τ^{on} stopped. Moreover, its residual-capacity and current-allocation resources cannot increase after that time. By the definition of R_τ, every feasible combination of such paths has total augmentation at most R_τ. The same argument shows that no length-one path ending at an offline contract can be created after the offline stage.

Now consider the alternating-path decomposition between the maintained allocation and an allocation attaining $A_t^\star$. Let $\mathcal{C}$ be the resulting set of paths, and let Δ_C denote the augmentation amount along each $C \in \mathcal{C}$. Then

$$A_t^\star - A_t = \sum_{C \in \mathcal{C}} \Delta_C.$$

Since no path with $\ell_C = 0$ remains and the paths with $\ell_C = 1$ are bounded by the corresponding remaining online reallocation chains,

$$\sum_{\substack{C \in \mathcal{C} \\ \ell_C = 1}} \Delta_C \le \sum_{\tau=1}^{t} R_\tau.$$

Every other path satisfies $\ell_C \ge 2$. Since such a path removes Δ_C from ℓ_C current allocation edges, and the total amount removed from any edge cannot exceed its current allocation,

$$\sum_{\substack{C \in \mathcal{C} \\ \ell_C \ge 2}} \ell_C \Delta_C \le A_t.$$

Therefore,

$$\begin{aligned} 2(A_t^\star - A_t) &= 2 \sum_{\substack{C \in \mathcal{C} \\ \ell_C = 1}} \Delta_C + 2 \sum_{\substack{C \in \mathcal{C} \\ \ell_C \ge 2}} \Delta_C \\ &\le 2 \sum_{\tau=1}^{t} R_\tau + \sum_{\substack{C \in \mathcal{C} \\ \ell_C \ge 2}} \ell_C \Delta_C \\ &\le 2 \sum_{\tau=1}^{t} R_\tau + A_t. \end{aligned}$$

Rearranging gives

$$A_t \ge \frac{2}{3} \left(A_t^\star - \sum_{\tau=1}^{t} R_\tau \right).$$

If every arrival is processed until no admissible online reallocation chain remains, then $R_\tau = 0$ for every $\tau \le t$. Hence,

$$A_t \ge \frac{2}{3} A_t^\star.$$

□